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Embracing the digital shift: Leveraging AI to foster employee well-being and engagement in remote workplace settings in the Asia Pacific region

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ABSTRACT

The COVID-19 pandemic has accelerated the shift to remote work, transforming working patterns and presenting new challenges for employees. As remote work continues to grow, organizations must adapt to ensure adequate performance and employee well-being. This study pioneered investigations of the role of artificial intelligence (AI) in enhancing work engagement and well-being in remote work settings. Drawing on the socio-technical system (STS) theory, we examine the mediating effects of human-AI teaming, task-technology fit, and perceived e-leadership support on the relationship between AI digital assistant skills and employee well-being. Our quantitative study of 360 sampling subjects reveals a significant positive relationship between AI digital assistant skills and human-AI teaming, task-technology fit, and perceived e-leadership support. Furthermore, these variables significantly influence employee well-being, which in turn has a positive effect on work engagement. Our findings underscore the importance of prioritizing human aspects within AI adoption, advocating for a balanced approach that enhances organizational efficiency while promoting positive workforce outcomes. This study highlights the need to consider the human factor when implementing technology in remote work environments, emphasizing the importance of a people-centric approach to AI adoption.
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1. Introduction

The transition from the industrial to the digital age has reshaped organizational structures, emphasizing flexibility and adaptability. Remote work, a key outcome of this shift, has become a significant aspect of modern work culture, driven by advancements in digital technology and evolving attitudes toward work locations (Bentley et al., 2016). The COVID-19 pandemic further accelerated this shift, making remote work a prevalent practice (Contreras, Baykal, & Abid, 2020; Contreras et al., 2020; Dogra & Priyashantha, 2023). Remote work allows employees to collaborate from locations outside traditional office settings, leveraging information and communication technology (ICT) for effective communication (Aguilar-Rodríguez, Bernal-Torres, Artieda-Cajilema, & Tapia-Andino, 2023). While remote work offers numerous benefits, including cost savings and increased access to global markets (Kilcullen, Feitosa, & Salas, 2022), it also presents challenges such as communication barriers, coordination difficulties, and personal issues like motivation and anxiety (Winter, 2020).

The rise of AI has further transformed the workplace. AI technologies are increasingly integrated into organizations, offering potential enhancements in productivity and operational efficiency (Woolley, Gupta, & Glikson, 2024). AI's impact is substantial, with predictions suggesting significant economic growth and the creation of new job opportunities (PWC, 2020; WEF, 2018). However, the integration of AI in the workplace also poses challenges, particularly in addressing the needs and well-being of human workers. There is a growing concern that AI implementation often overlooks human-centric perspectives, potentially leading to job displacement and diminished job satisfaction (Berretta, Tausch, Ontrup, et al., 2023).

This study explores integrating AI technologies, particularly AI-based digital assistants, in remote work settings. It examines how AI can influence employee well-being and work engagement, focusing on human-AI teaming, task-technology fit, and perceived e-leadership support as mediating factors. The study is grounded in the sociotechnical system (STS) theory, which emphasizes the interplay between organizational social and technological

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components (Appelbaum, 1997). This STS perspective is proposed as essential to examine the role of humans and technology in affecting both instrumental and humanistic outcomes in the fourth industrial revolution (Melville, Robert, & Xiao, 2023). As we all experience human and AI teaming by involving more advanced AI digital assistants in the workplace, this paper's first research objective is to explore human adaptive behavior.

By evaluating these interactions between human-AI teaming perceived by human workers, the second research objective of the study is to illustrate the insights into how organizations can effectively integrate AI while promoting employee positive outcomes as the perceived level of well-being and work engagement. As Parent-Rocheleau and Parker (2022) point out the work design drawing on STS is essential in moderating work performance, including perceived well-being and performance. We test the consequences on well-being and work engagement, are also scarce in the literature but is important in work design, especially in remote work environments, while workers freely choose their work layouts and environments.

Therefore, the research questions guiding this study are: (1) RQ1: How does integrating AI digital assistants impact employees' well-being in remote work settings? (2) RQ2: How do human-AI teaming, task-technology fit, and perceived e-leadership support mediate the relationship between AI digital assistants' integration and employees' well-being? (3) RQ3: How does the overall impact of AI digital assistant skills on employees' well-being influence employees' work engagement in remote work settings?

2. Theoretical background and research framework

2.1. Remote work

Remote work, broadly defined, encompasses various forms of employment outside the traditional office, utilizing information and communication technologies (ICTs). Tietze & Musson, (2005) anticipated these shifts decades ago, predicting a work paradigm characterized by flexibility, mobility, and technological mediation. Historical antecedents, such as the oil crisis in the 1970s, also catalyzed the early adoption of telecommuting as an alternative to physical commuting (Nilles, 1976).

The rise of remote work has been driven by technological advancements, particularly in ICT infrastructure, which enabled seamless communication and file-sharing capabilities, thus supporting the relocation of work processes to various geographic locales (Messenger & Gschwind, 2016). This work mode offers notable benefits, including enhanced job satisfaction, reduced work-family conflict, and improved work-life balance (Othman, Yusef, & Osman, 2009, pp. 3–15; Vega, Anderson, & Kaplan, 2015). However, the success of remote work is contingent upon robust support systems, including management, peer, and technological backing (Bentley et al., 2016).

Moreover, remote work contributes positively to corporate sustainability efforts by reducing the environmental impact of daily commuting and optimizing resource use (Giovanis, 2018). The flexibility inherent in remote work arrangements also promotes more excellent employment opportunities, particularly for women and individuals with disabilities, by accommodating diverse personal and professional needs (Morgan, 2004).

Despite these advantages, remote work is not without challenges. Social isolation, reduced opportunities for workplace learning, and heightened anxiety about career prospects are significant concerns (Fedáková & Ištoňová, 2017; Maruyama & Tietze, 2012). Additionally, remote work may lead to work intensification, diminished employer trust, and ethical issues related to employee exploitation and privacy intrusions (Cortellazzo, Bruni, & Zampieri, 2019; Kaplan, Engelsted, Lei, & Lockwood, 2018). These factors necessitate a balanced approach to remote work implementation, ensuring its flexibility does not come at the cost of increased employee stress and reduced organizational cohesion (Putnam, Myers, & Gailliard, 2014).

2.2. Role of AI digital assistant

Artificial Intelligence (AI) has emerged as one of the most transformative technologies of the 21st century. Defined as systems capable of mimicking human intelligence with minimal human intervention (Haenlein & Kaplan, 2019), AI has significantly influenced organizational processes, driven business model innovation, and enhanced operational efficiency (Sam Ransbotham, & KironLafountainKhodabandeh, 2021) and emotional intelligence in business administration (Yeke, 2023). Despite concerns about AI's impact on employment and potential ethical challenges, its role in augmenting human productivity remains paramount (Calo, 2015; Singh & Shaurya, 2021).

In remote work environments, AI technologies, such as AI-mediated communication tools, enhance team interactions by reducing misunderstandings and facilitating conflict resolution, particularly in culturally diverse virtual teams (Hancock, Naaman, & Levy, 2020). Furthermore, AI-driven management systems have been shown to increase employee engagement through practical guidance and monitoring (Hughes, Robert, Frady, & Arroyos, 2019).

One prominent application of AI in the workplace is through digital assistants, which have evolved from early systems like Eliza,[1] which was developed by Joseph Weizenbaum in 1966, to sophisticated tools used by global IT giants such as Google and Amazon. These AI-powered assistants, including chatbots and voice-activated systems, are now integral to modern workplace communication platforms like Microsoft Teams and Slack, enhancing productivity by automating routine tasks (Maedche et al., 2019). Gartner (2019) predicted that by 2021, 25% of digital workers would regularly use virtual assistants, highlighting their growing importance.

AI digital assistants integrate individual user objectives with advanced technologies such as natural language processing, machine learning, and computer vision (Maedche et al., 2019; Russell & Norvig, 2016). These systems facilitate task completion by processing input through conversational interfaces and generating relevant outputs. However, the effective utilization of AI in the workplace requires employees to develop AI literacy, including data analysis, decision-making skills, and understanding AI's potential biases (Hermann, 2022; Long & Magerko, 2020). Despite the challenges, AI digital assistants offer significant benefits, including improved efficiency and enhanced workplace communication, positioning them as critical tools in the future of work.

2.3. Socio-technical system (STS) theory

The Socio-Technical System (STS) theory, first introduced by Trist (1981), seeks to understand the interaction and mutual shaping of a system's social and technological elements. It emphasizes the concept of joint optimization, where the design of work systems must balance both social and technical components to achieve favorable outcomes. This approach differs from traditional methods prioritizing technical design, often leading to suboptimal performance and significant social costs. STS theory also incorporates boundary management, which protects the work

[1] Eliza: https://cse.buffalo.edu/∼rapaport/572/S02/weizenbaum.eliza.1966.pdf (Data is accessed on 2024.8).

system from external disruptions while ensuring necessary information and resource flow (Appelbaum, 1997).

STS design advocates for work organization that aligns with company goals, typically through participative approaches involving employees in task creation. Flexibility in work design is also crucial, allowing for swift adaptation to technological changes and enhancing operational efficiency. According to Trist (1981), the interplay between social and technical subsystems is central to STS, with the social system encompassing organizational culture, roles, communication styles, and interpersonal relationships, while the technical system includes tools and processes required for task completion, as shown in Fig. 1.

In remote work, technology has significantly influenced the shift to digital workspaces, where the interaction between humans and machines becomes more prominent. Lannotta, Meret, and Marchetti (2020) identify three key aspects of this transformation: behavior modification, shared meaning in change management, and the combination of technology-mediated and in-person interactions. Additionally, Rapaccini, Saccani, Kowalkowski, Paiola, and Adrodegari (2020) highlight the three pillars of this disruption: the technical dimension (digital technologies enabling remote work), the social component (organizational behaviors and HR management), and the physical dimension (workspace organization and ergonomic support).

The STS framework, integrating technological and social subsystems, serves as an ideal guide for interpreting the shift to digital workspaces. Technological advancements such as AI, the Internet of Things, and quantum computing have reshaped work tasks and the interconnection between humans and machines (Autor, Dorn, Katz, Patterson, & Van Reenen, 2020). The social subsystem focuses on human factors like motivation, personality, and the structure of interpersonal connections within the organization, which are critical for managing creative and non-linear thinking in remote work environments.

2.4. AI digital assistant skills of employees in remote settings

Proficiency with digital tools and resources, including AI, is one of the eight critical abilities for lifelong learning (Van Laar, Van Deursen, Van Dijk, & De Haan, 2017). As AI skills are developed at the individual level, they contribute to the collective AI competence of employees, which constitutes an organizational capability (Long & Magerko, 2020). This organizational capability facilitates human-AI teaming, where employees can effectively interact and cooperate with AI technologies. According to Hermann (2022), the ability to interact and collaborate with AI efficiently is essential for creating AI skills. These competencies enable workers to understand AI's potential and limitations, thereby aiding in comprehending how AI can be used to solve specific business challenges (Pillai & Sivathanu, 2020). Thus, employees who possess strong AI digital assistant skills are better equipped to team with AI systems, enhancing productivity and business process re-engineering in remote work settings. Based on this, we hypothesize that.

H1. AI digital assistant skills of employees positively influence human-AI teaming in remote work settings.

Utilization skills, which encompass the know-how required to decipher, clarify, and grasp the results produced by AI-based decision support systems, are crucial for improving data-driven decision-making in various corporate situations (Chowdhury, Budhwar, Dey, Joel-Edgar, & Abadie, 2022). These domain-specific knowledge-based competencies ensure that employees can use AI tools effectively, enhancing task-technology fit. The ability to understand and leverage AI technologies aligns the tasks with the technological capabilities available, ensuring that employees can perform their roles efficiently and effectively. Organizations can optimize their resources and business processes by developing these skills, as AI skills provide a strategic advantage that is difficult for other firms to replicate (Mikalef & Gupta, 2021). Consequently, employees' AI digital assistant skills play a significant role in achieving a high task-technology fit in remote work settings. Therefore, we developed a hypothesis as follows.

H2. AI digital assistant skills of employees positively influence task-technology Fit in remote work settings.

We hypothesize that the AI skills of employees positively influence perceived e-leadership support. This relationship is grounded in leaders' active awareness of ICTs, which includes staying abreast of technological innovations, pursuing technological solutions to enhance organizational efficiency, and critically evaluating the value of these technologies (Van Wart, Roman, Wang, & Liu, 2017). Leaders who demonstrate a proactive approach to integrating AI foster an environment where employees are encouraged to develop and utilize their AI skills. When leaders actively know and support AI technologies, they create a culture that values innovation, making employees more likely to feel supported in their AI endeavors. Consequently, employees perceive more outstanding e-leadership support, as they see their leaders endorsing AI initiatives and facilitating their technological growth and development. Therefore, we propose the hypothesis as follows.

H3. AI digital assistant skills of employees positively influence perceived e-leadership support in remote work settings.

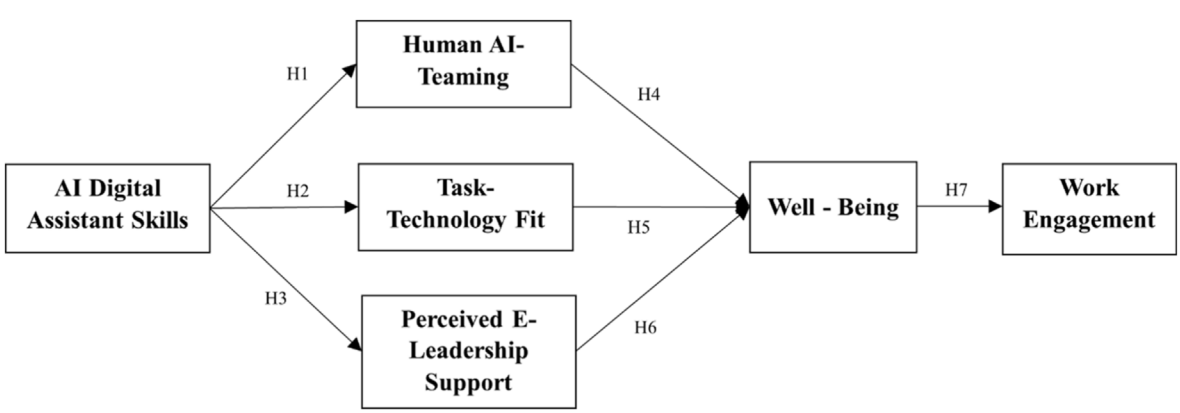

Fig. 1. Research framework.

2.5. Human-AI teaming in remote settings

Human-AI teaming positively influences well-being in remote work settings by fostering a collaborative environment where AI is viewed as an active partner rather than just a tool. The CASA paradigm suggests that individuals perceive autonomous AI-driven technologies as social entities, leading to a more favorable outlook on collaboration (Rix, 2022). This shift allows employees to delegate monotonous and safety-critical tasks to AI, freeing them to focus on more creative and identity-forming work, enhancing job satisfaction and overall well-being (Berretta, Tausch, Peifer, & Kluge, 2023). Furthermore, AI and other technologies help employees achieve challenging goals and deliver superior service quality, which boosts engagement and productivity (Li, Yin, Qiu, & Bai, 2021). To fully leverage these benefits, employees must meet specific conditions such as understanding AI behavior, building reliable rapport with AI systems, and continuously upskilling to adapt to evolving roles (Zirar, 2023). These factors collectively contribute to a positive impact on employee well-being in remote work settings. Hence, we propose this hypothesis.

H4. Human-AI teaming positively influences well−being in remote work settings.

2.6. Task-technology fit in remote settings

Task-technology fit positively influences well-being in remote work settings by ensuring that technology aligns with the specific requirements of tasks and the capabilities of users. According to the Task-Technology Fit (TTF) theory, performance improves, and tasks are completed more efficiently when technology, task requirements, and individual skills are well-matched (Goodhue & Thompson, 1995). In remote work settings, where reliance on technology for communication and collaboration is crucial, technologies such as Zoom, Microsoft Teams, and Skype must be designed to meet teleworking needs effectively (Ameen, Papagiannidis, Hosany, & Gentina, 2023). Properly designed technology that provides real-time, delay-free services, ensures secure and reliable operations and includes features tailored to remote work enhances productivity and job satisfaction while reducing technostress (Dolgui, Bernard, Lemoine, Von Cieminski, & Romero, 2021). Conversely, poor task-technology fit can lead to underutilization or bypassing of technology, which may increase technostress and diminish well-being. Therefore, ensuring a good task-technology fit enhances employee well-being in remote work settings. Thus, we proposed the hypothesis as the following.

H5. Task-technology fit positively influences well−being in remote work settings.

2.7. Perceived E-leadership support in remote settings

Perceived e-leadership support positively influences well-being in remote work settings by ensuring leaders effectively utilize traditional and digital communication channels to foster a supportive and engaging environment. E-leadership requires a blend of advanced technological skills, clear communication, and the ability to handle virtual interactions seamlessly (Van Wart, Roman, Wang, & Liu, 2019). Effective e-leaders build environments that encourage openness, participation, and autonomy, reducing the need for constant monitoring and enhancing employee comfort and satisfaction (Cortellazzo et al., 2019). By leveraging technology to maintain clear and error-free communication and creating an e-social environment that mitigates isolation, e-leaders contribute to increased employee engagement and well-being (Roman et al., 2019). Moreover, successful e-leaders manage organizational change and promote innovation, aligning with strategic objectives while supporting team development and productivity (Schwarzmüller, Brosi, Duman, & Welpe, 2018). Consequently, positive perceptions of e-leadership support led to improved well-being by fostering a cohesive, productive, and supportive remote work environment. Therefore, we proposed this hypothesis.

H6. Perceived E-leadership support positively influences well−being in remote work settings.

2.8. Employees' well-being and work engagement

The well-being of employees positively influences their work engagement in remote settings by fostering a more fulfilling and satisfying work experience. Research shows that employees who experience higher levels of psychological well-being are more likely to exhibit greater job engagement, commitment, and performance (Singh, David, & Mikkilineni, 2018; Wang, Li, & Li, 2017). Elevated well-being enhances employees' ability to view their work positively, which in turn boosts their engagement levels and dedication to their tasks (Tadić, Bakker, & Oerlemans, 2015). For example, studies have demonstrated a significant correlation between psychological well-being and work engagement among various groups, including hotel employees and social workers (Aiello & Tesi, 2017; Gyu Park, Sik Kim, Yoon, & Joo, 2017; Tesi, Aiello, & Giannetti, 2019). In remote work settings, where employees face unique challenges such as isolation and blurred work-life boundaries, maintaining high well-being is crucial for sustaining engagement and productivity. Thus, organizations prioritizing and supporting employee well-being will likely see enhanced work engagement, resulting in improved performance and organizational success. In light of this, we proposed the hypothesis as the following.

H7. The well-being of employees positively influences their engagement in remote work settings.

3. Research method

3.1. Sample and procedure

A quantitative survey research design was adopted for the investigation. Before the survey's widespread dissemination, a pilot test was conducted to assess the survey's effectiveness and comprehensibility. This pilot study lasted one week in February and had 30 participants. The questionnaire was improved based on feedback from the pilot test, confirming the meanings and reducing the ambiguities in the text. The following factors were taken into consideration when choosing the survey sample: (1) Participants must be currently working remotely, either fully or in a hybrid model; (2) Participants must use AI digital assistants for work purposes; (3) Participants must be employed by a company, excluding freelancers. In this study, all respondents were based in Indonesia. The survey was disseminated through social media platforms (Twitter, Instagram, Facebook, LinkedIn) and remote work communities via WhatsApp and Discord to reach a representative sample. Data collection occurred from May 6th to May 29th. Data were initially collected from 360 respondents via social media. After screening, 74 responses were excluded due to non-eligibility or data quality issues, resulting in a final sample of 286 respondents. The majority (62.9%) were aged 25−34, with a slight gender skew towards females (56.3%). All respondents were Indonesian nationals, with 58.7% working fully remotely and 41.3% in hybrid setups. Most respondents were employed in corporations (42%), predominantly in business analyst roles (50%). Organizations with fewer than 50 employees constituted 44.1% of the sample,

with the technology sector being the most represented (29%). In terms of job positions, 42.3% were at the entry-level. Experience with AI ranged from under one year to over a decade, with ChatGPT being the most commonly used tool (50.9%). AI tools were employed for scheduling, idea generation, automating repetitive tasks, and information retrieval as the most frequent ones (see *Appendix A* for details).

3.2. Measurement scales

The measurement tools employed in this study were drawn from earlier academic publications. Respondents answered 33 items on a 7-point Likert scale, ranging from "strongly disagree" to "strongly agree." To assess AI digital assistant skills and their integration in the workplace, 5 relevant items were selected from Chowdhury et al. (2022). Human-AI collaboration was evaluated using 5 curated items from the JOPI (Job Perception Intervention) scale by Berrett et al. (2023), focusing on the collaborative relationship between humans and AI, where AI manages routine tasks, allowing humans to engage in more expressive work. Perceived e-leadership support was measured using 5 items—4 from Elyousfi, Anand, and Dalmasso (2021), covering trust, communication, collaboration, and behavior in virtual teams, and 1 from Liu et al. (2018), assessing ICT usage as an indicator of effective e-leadership. The well-being of working with AI was assessed using 5 items, including 3 from Berretta, Tausch, Peifer, and Kluge (2023) on the impact of AI on well-being, and 2 adapted from Diener's Psychological Well-Being Scale, tailored to the context of AI in the workplace. Finally, work engagement was measured using 8 items from Marikyan, Papagiannidis, Rana, Ranjan, and Morgan (2022), evaluating employees' commitment and motivation in AI-integrated remote work settings. This comprehensive measurement approach ensures a thorough evaluation of the critical constructs relevant to AI integration in remote work environments.

3.3. Data analysis

This study used structural equation modeling (SEM) as the data analysis method. Two steps were conducted. First, we used a confirmatory factor analysis (CFA) method to study a measurement model to evaluate the measurement quality of constructs. Second, in order to verify the study hypotheses, we estimated a structural model (Hair et al., 2019). A comparison of the model fit indices between the original and modified measurement models is reported in *Appendix B*, where low factor loading items were removed. Item WE8 was deleted due to its low factor loading of 0.16, while the others were retained. Both models show reasonable fit, with Chi-square/df ($2.482 < 3.00$), RMSEA ($0.072 < 0.08$), and P-value ($0.000 < 0.05$) meeting acceptable thresholds. However, GFI, AGFI, NFI, and CFI did not reach the desired levels, indicating the need for further refinement. Deleting low-loading items led to slight improvements in some indices, but overall model fit remained largely unchanged.

Appendix C highlights key statistical indicators. Composite Reliability (CR) values for all constructs exceed 0.7, confirming strong reliability. The standardized loadings, all statistically significant ($p < 0.001$) with high t-values, indicate that the indicators effectively measure their respective constructs. Discriminant validity is evaluated by comparing Maximum Shared Variation (MSV) to Average Variance Extracted (AVE), with most constructs showing acceptable discriminant validity as their MSV values are lower than their AVE. However, Human-AI Teaming (HAT) exhibits higher MSV values, suggesting potential issues. Most constructs have AVE values above 0.5, though HAT and Perceived E-Leadership Support (ELS) fall slightly below, indicating room for improvement in these constructs' measurement items.

The Harman's single-factor test showed that the first component explained 33.275% of the variance, which is well below the 50% threshold that suggests strong Common Method Bias (CMB) (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). This indicates that CMB is not a concern in this study, ensuring the validity and reliability of the results (See *Appendix E* for details).

3.4. Result

Fig. 2 explores the impact of AI digital assistant skills, human-AI teaming, task-technology fit, and perceived e-leadership support on outcomes in remote work. As shown in Table 1. AI digital assistant skills significantly enhance human-AI teaming ($\beta = 0.605$), task-technology fit ($\beta = 0.573$), and perceived e-leadership support ($\beta = 0.502$), all with p-values <0.001, indicating their crucial role in effective collaboration with AI, improving productivity, communication, and coordination. Human-AI teaming positively affects employees' well-being ($\beta = 0.239$), while task-technology fit ($\beta = 0.448$) and perceived e-leadership support ($\beta = 0.247$) also contribute to better well-being. Additionally, employees' well-being strongly influences work engagement ($\beta = 0.551$). These results underscore the importance of AI in fostering a healthy remote work environment. Statistical evidence strongly supports each hypothesis, highlighting the critical factors for successful remote work.

Table 2 explores the mediating effects of human-AI teaming (HAT), task-technology fit (TTF), and perceived e-leadership support (ELS) on the relationship between AI digital assistant skills (SKILL) and employees' well-being (WB) in remote work settings. The findings indicate full mediation for all three mediators, as the indirect effects of AI skills on well-being through HAT, TTF, and ELS are statistically significant ($p < 0.001$), while the direct effects are not significant ($p = 0.220$). This suggests that the positive impact of AI skills on well-being is fully mediated by improved human-AI teaming, better task-technology fit, and enhanced perceived e-leadership support. These mediating factors fully explain the relationship between AI capabilities and employee well-being, underscoring their critical role in promoting well-being in remote work environments.

4. Conclusions and discussions

4.1. Research discussion

This study underscores the critical role of AI digital assistant skills in enhancing human-AI teaming, task-technology fit, and perceived e-leadership support, all of which contribute to improved employee well-being and work engagement in remote settings. The findings strongly support the hypothesis that AI digital assistant skills positively influence human-AI teaming (H1), with evidence showing that employees proficient in these tools experience smoother interactions and more cohesive teamwork ($\beta = 0.605$, $t = 7.571$, $p < 0.001$). This aligns with the socio-technical system (STS) theory, suggesting that enhancing employees' AI-related skills can significantly improve socio-technical alignment, leading to better organizational outcomes (Abulsaoud Ahmed Younis & Adel, 2020; Chowdhury et al., 2022).

Similarly, our results confirm that AI digital assistant skills positively affect task-technology fit (H2) ($\beta = 0.573$, $t = 7.242$, $p < 0.001$), highlighting the importance of aligning technology with task requirements to optimize performance and reduce cognitive load (Goodhue & Thompson, 1995). This finding is consistent with the Task-Technology Fit (TTF) model, which asserts that performance improvements are realized when technology effectively

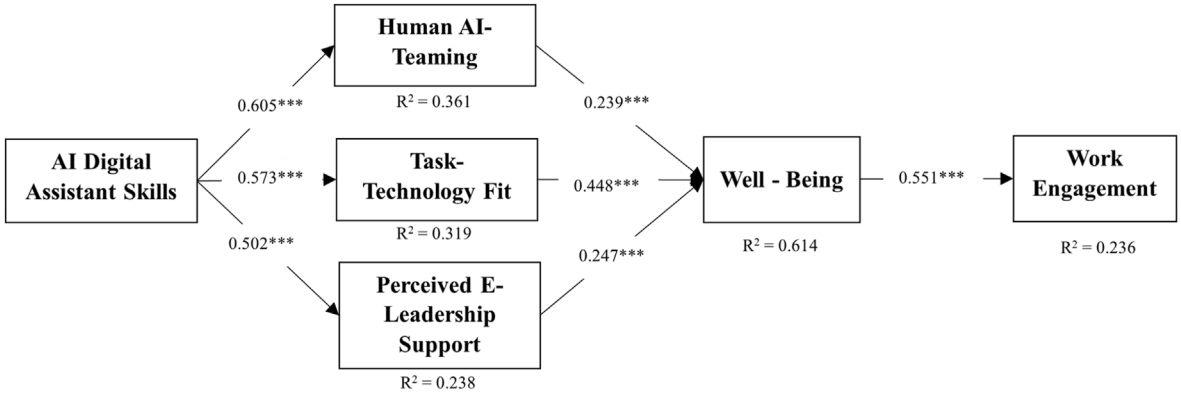

Fig. 2. Results of path analysis.

Table 1
Results of path analysis.

Hypotheses	Standardized Coefficient	t-value	p-value	Results
H1: AI digital assistant skills of employee positively influence human-AI teaming in remote work settings.	0.605	7.571	[a]	Supported
H2: AI digital assistant skills of employee positively influence task-technology fit in remote work settings.	0.573	7.242	[a]	Supported
H3: AI digital assistant skills of employee positively influence perceived E-leadership support in remote work settings.	0.502	6.404	[a]	Supported
H4: Human-AI teaming positively influences well–being in remote work settings.	0.239	6.897	[a]	Supported
H5: Task-technology fit positively influences well-being in remote work settings.	0.448	4.040	[a]	Supported
H6: Perceived E-leadership support positively influences well–being in remote work settings.	0.247	3.570	[a]	Supported
H7: The well-being of employees positively influences their engagement in remote work settings.	0.551	6.052	[a]	Supported

Notes: *p < 0.05 and they were significant at a t-value >1.96.
[a] $p < 0.001$.

Table 2
Result of mediating effects and path analysis.

Mediating Effects and Path Analyses	Direct Effect	Indirect Effect	Confidence Interval		P-Value	Results
			Lower Bound	Upper Bound		
Well – Being						
SKILL - > HAT - > WB	0.098 (0.220)[N.S]	0.272	0.167	0.426	0.000	*Full mediation*
SKILL - > TTF - > WB		0.095	0.004	0.306	0.035	*Full mediation*
SKILL - > ELS - > WB		0.086	0.032	0.177	0.003	*Full mediation*
Work Engagement						
SKILL - > HAT - > WE	0.210 (0.027)[a]	0.105	0.007	0.225	0.037	*Partial-mediation*
SKILL - > TTF - > WE		0.033	−0.053	0.157	0.404	*No mediation*
SKILL - > ELS - > WE		0.197	0.106	0.339	0.000	*Partial mediation*

Notes.
[a] $p < 0.05$ and they were significant at a t-value >1.96.

supports task requirements.

Moreover, the study shows that AI digital assistant skills enhance perceived e-leadership support (H3) ($\beta = 0.502$, $t = 6.404$, $p < 0.001$). Technological competence appears to improve social dynamics within organizations, as effective use of AI tools can lead to better communication and morale (Van Wart et al., 2019). E-leadership, which relies heavily on digital communication, benefits from employees' proficiency in AI tools, leading to more effective and supportive leadership.

The study further reveals that human-AI teaming significantly enhances employee well-being (H4) ($\beta = 0.239$, $t = 6.897$, $p < 0.001$). This outcome can be attributed to the seamless integration of technical and social systems, which fosters a supportive work environment that promotes well-being (Berretta, Tausch, Peifer, & Kluge, 2023). The mediation analysis indicates that human-AI teaming fully mediates the relationship between AI skills and well-being (Indirect Effect = 0.272, $p < 0.001$), underscoring the importance of collaborative intelligence in achieving better outcomes (Jarrahi, 2018).

Our findings also show that task-technology fit significantly improves employee well-being (H5) ($\beta = 0.448$, $t = 4.040$, $p < 0.001$), reinforcing the need for a harmonious balance between technological capabilities and job requirements. Proper alignment between these elements can enhance efficiency, reduce techno-stress, and promote employee well-being (Ameen et al., 2023). The mediation analysis supports that task-technology fit fully mediates the relationship between AI skills and well-being (Indirect Effect = 0.095, $p = 0.035$), emphasizing the importance of matching technology to tasks to foster employee well-being.

Our study shows that perceived e-leadership support significantly enhances employee well-being ($\beta = 0.247$, $t = 3.570$, $p < 0.001$). Effective e-leadership mitigates remote work challenges

by offering essential guidance and emotional support, boosting engagement and well-being (Bentley et al., 2016; Contreras et al., 2020). Additionally, e-leadership fully mediates the relationship between AI skills and well-being (Indirect Effect = 0.086, $p = 0.003$), highlighting that AI proficiency improves perceived leadership support, aligning with previous findings on the importance of effective leadership in remote settings (Arokiasamy & Tat, 2020; Ibrahim, Yaakob, & Yusof, 2018).

Lastly, our study confirms the positive correlation between well-being and work engagement ($\beta = 0.551$, $t = 6.845$, $p < 0.001$), consistent with existing literature that highlights the crucial role of well-being in boosting engagement (Singh et al., 2018; Teoh Yi Zhe & Keikhosrokiani, 2021; Wang et al., 2017). Prioritizing employee well-being is essential for sustaining high levels of work engagement, especially in remote work environments.

4.2. Theoretical contribution

Using the socio-technical systems (STS) theory, our study adds numerous theoretical perspectives to the understanding of technology uptake and integration inside businesses. By examining the effects of AI digital assistant skills on various mediating variables such as human-AI teaming, task-technology fit, and perceived e-leadership support, we extend the STS framework to encompass contemporary technological advancements and their influence on organizational outcomes. It highlights the synergy between AI digital assistant skills and human capabilities, optimizing both social and technical components in remote work settings (Klaser, Cuel, & Casari, 2023). Our human-centric approach emphasizes collaborative intelligence, where AI enhances rather than replaces human capabilities (Berretta, Tausch, Ontrup, et al., 2023). We demonstrate that AI skills improve human-AI collaboration, supporting better task performance and well-being (Woolley et al., 2024). Additionally, AI skills can also improve task-technology fit and mediates the relationship between AI skills and employee well-being, underscoring the importance of well-matched technology for reducing cognitive load and boosting efficiency (Ameen et al., 2023). Our findings also extend the role of e-leadership in remote contexts, showing that proficiency in AI tools enhances communication and support, thereby improving morale and engagement (Contreras et al., 2020). Overall, our study provides a comprehensive view of how AI can enhance employee experiences and organizational outcomes through a balanced socio-technical approach (Singh et al., 2018).

4.3. Managerial implication

Our study provides key insights for integrating AI digital assistants to enhance organizational outcomes. Managers should focus on AI skills training to improve human-AI collaboration and efficiency, ensuring employees are adept at using evolving technologies. AI tools should be designed with a user-centric approach, incorporating advanced natural language processing and ethical guidelines to ensure seamless integration and address privacy concerns. Encouraging human-AI teaming can reduce cognitive load and stress by automating routine tasks and supporting decision-making. Regular assessments of task-technology fit and employee feedback are crucial for maintaining alignment between technology and job requirements. Additionally, effective e-leadership is vital for remote work settings, where managers should use digital tools to provide consistent guidance, resources, and emotional support, ensuring clear communication and a supportive work environment.

4.4. Limitations and future research opportunities

This study has several limitations. Firstly, its focus on Indonesia may restrict the generalizability of the findings to other cultural and geographic contexts, as cultural influences significantly impact AI adoption and use. Future research should investigate AI digital assistant skills across diverse cultures and regions to understand how cultural factors affect AI integration. Secondly, the mixed sample of hybrid and fully remote workers may have influenced the results, as these groups face different challenges. Future studies should separately analyze these work modes to develop targeted strategies. Additionally, the reliance on self-reported survey data introduces biases such as memory and social desirability, suggesting the need for mixed methods that include qualitative data for deeper insights. Lastly, the cross-sectional design limits causal inferences; longitudinal studies are needed to explore the evolving effects of AI integration on employee engagement and well-being over time.

Appendix A

Table A-1
Demographic information of the sampling subjects.

Items	Type	N	%
Remote Work Type	Fully remote	168	58.7
	Hybrid	118	41.3
Job Role	Digital Marketer	36	12.6
	Data Scientist	16	5.6
	Business Analysts	143	50.0
	Software Developer	13	4.5
	Graphic Designer	54	18.9
	Others (Translator, Virtual Assistant, Human Resource Staff, Product Manager, Project Manager, UI/UX Designer, etc.)	24	8.4
Age	18–24	60	21.0
	25–34	180	62.9
	35–44	37	12.9
	45–54	9	3.1
Gender	Male	125	43.7
	Female	161	56.3
Organization Type	Corporation	120	42.0
	Small Medium Enterprise	24	8.4
	Startup	95	33.2

(*continued on next page*)

Table A-1 (*continued*)

Items	Type	N	%
	Non-Profit	9	3.1
	Others (Agency, Community, University, Government)	38	13.3
Number of Employees	Less than 50	126	44.1
	50–250	81	28.3
	251–1000	37	12.9
	More than 1000	42	14.7
Sector	Manufacturing	19	6.6
	Technology	83	29.0
	Healthcare	13	4.5
	Finance	17	5.9
	Retail	13	4.5
	Education	39	13.6
	Services	53	18.5
	Others (Entertainment, Hospitality, Real Estate, Information Media)	49	17.1
Job Position Level	Entry Level	121	42.3
	Intermediate of Experienced (Senior Staff)	107	37.4
	First Level Management	27	9.4
	Middle Management	31	10.8
Period of Working with AI	Less than one year	138	48.3
	1–3 years	138	8,3
	4–6 years	9	2.8
	7–10 years	1	0,3
	More than ten years	1	0,3
AI Types for Work	ChatGPT	266	50.9
	Google Assistant	108	20.7
	Amazon Alexa	6	1.1
	Microsoft Cortana	17	3.3
	HubSpot	15	2.9
	Data Robot Paxata	2	0.4
	Tableau	18	3.4
	Adobe Sensei	5	1.0
	GitHub Copilot	43	8.2
	Others (Bard, Canva, Bixby, Claude, Gemini, Microsoft Copilot, Perplexity, ChatPDF, Siri, etc.)	43	8.2
Tasks Handled by AI Digital Assistants	Schedule Work Tasks	71	9.7
	Set a reminder	61	8.3
	Generate ideas	190	25.9
	Search information	216	29.4
	Automate repetitive tasks	54	7.3
	Optimizing performance	105	14.3
	Others (Brainstorming partner, coding, communicating with coworkers, taking notes, managing files, creating content, debugging problems, writing, fixing grammar, translation)	38	5.2

Appendix B

Table B
Goodness-of-fit Measures for the Measurement Model (CFA).

Criteria	Original Model	Delete Items with Low Factor Loadings
Chie-square/df < 3.00	2.412	2.482
GFI >0.90	0.803	0.805
AGFI >0.90	0.770	0.771
NFI >0.90	0.801	0.807
CFI >0.95	0.872	0.874
RMR <0.05	0.086	0.082
RMSEA <0.08	0.070	0.072
P-value <0.05	0.000	0.000

Appendix C

Table C
Analysis and interpretation of measurement model indicators.

Items		Construct	Standardized loading	Error variance	*p*	CR Composite Reliability	AVE
1. I have knowledge about AI digital assistants.	←	SKILL	0.760	0.086	***	0.866	0.566
2. I have relevant skills in using AI digital assistants in my work.	←		0.808	0.082	***		
3. I have the competencies to understand how AI digital assistants will execute.	←		0.789	0.087	***		
4. I have the skills to interpret the AI digital assistant outputs.	←		0.668	0.114	***		
5. I have the skills to prepare inputs for AI digital assistants.	←		0.727	–	***		
1. Working with the digital assistant relieves me of tasks I am less inclined to do.	←	HAT	0.624	–	***	0.829	0.495
2. The digital assistant takes on important complementary tasks that I would otherwise not be able to complete due to time constraints.	←		0.621	0.124	***		
3. When working together with the digital assistant, we complement each other ideally.	←		0.753	0.116	***		
4. Working with the digital assistant means I can concentrate fully on the essential aspects of my work.	←		0.799	0.140	***		
5. Working with the digital assistant makes me feel like I'm part of a real team.	←		0.703	0.153	***		
1. Whoever developed this AI system thought about how to support my tasks meaningfully.	←	TTF	0.683	–	***	0.848	0.528
2. Someone who *understands* my work was involved in developing this AI system.*	←		0.773	0.126	***		
3. Someone who *knows* my work processes was involved in developing this AI system.	←		0.692	0.118	***		
4. AI system fits very well with the tasks I do.	←		0.738	0.105	***		
5. king with the AI system is an enrichment for my work.	←		0.744	0.105	***		
1. conversations between my supervisor and me were smooth when talking about work tasks.	←	ELS	0.661	0.081	***	0.832	0.499
2. as sure I could openly communicate my feelings to my supervisor.	←		0.786	0.108	***		
3. as confident that I could share my work difficulties with my supervisor.	←		0.752	0.118	***		
4. supervisor showed a willingness to take initiative.	←		0.693	0.112	***		
5. supervisor regularly used ICTs for leadership purposes. (*Ex*, utilize *project management software to assign tasks, conduct virtual meetings through video conferencing platforms, etc.*)	←		0.629	–	***		
1. I feel comfortable working with the AI system.	←	WB	0.717	–	***	0.866	0.566
2. I can imagine that working with such an AI system will be fun for me in the long run.	←		0.844	0.092	***		
3. I can imagine that working with such an AI system will help me in the long run.	←		0.838	0.085	***		
4. Working With AI, I lead a purposeful and meaningful life.	←		0.640	0.114	***		
5. With AI, I am competent in the activities that are important to me.	←		0.700	0.097	***		
1. Using the AI digital assistant for my work-related tasks has energized me at work.	←	WE	0.746	–	***	0.920	0.622
2. Using the AI digital assistant for my work-related tasks has made me feel strong and vigorous at work.	←		0.861	0.084	***		
3. Using the AI digital assistant to carry out my work-related tasks has made me enthusiastic about my job.	←		0.876	0.076	***		
4. Using the AI digital assistant to work out my work-related tasks has made me feel inspired by my job.	←		0.824	0.077	***		
5. Using the AI digital assistant for my work-related tasks has prepared me to work.	←		0.757	0.077	***		
6. Using the AI digital assistant to carry out my work-related tasks has made me happy when I am working intensely.	←		0.721	0.085	***		
7. Using an AI digital assistant to carry out my work-related tasks has made me proud of my work.	←		0.717	0.071	***		
8. Using the AI digital assistant to carry out my work-related tasks has made me feel immersed in my work *							

χ^2/d.f (97) = 2.482, GFI = 0.805, AGFI = 0.771, RMR = 0.082, p = 0.000.
Notes: $^{**}p < 0.01$.
*** $p < 0.001$.
* $p < 0.05$ and they were significant at a t-value >1.96. A regression weight was fixed at 1.00. CR=Composite Reliability; C.R=Critical Ratio (C.R = t-value); AVE = Average Variance Extracted. *: The item is deleted due to the low validity.

Appendix D

Table D
Evaluation of construct validity and reliability.

	CR	AVE	MSV	MaxR(H)	WE	SKILL	HAT	TTF	ELS	WE
WE	0.866	0.566	0.563	0.883	0.752					
SKILL	0.866	0.566	0.315	0.935	0.537	0.752				
HAT	0.829	0.495	0.563	0.952	0.750	0.525	0.704			
TTF	0.848	0.528	0.511	0.962	0.666	0.561	0.715	0.727		
ELS	0.832	0.499	0.356	0.968	0.552	0.440	0.581	0.434	0.707	
WE	0.920	0.622	0.356	0.978	0.446	0.494	0.528	0.457	0.597	0.788

Note: AVE: Average variance extracted, MSV: Maximum shared variance, MaxR(H)Maximum reliability.

Appendix E

Table E
Results of common method bias.

Total Variance Explained						
Factor	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	11.624	35.224	35.224	10.981	33.275	33.275
2	2.924	8.862	44.086			
3	2.237	6.780	50.866			
4	1.755	5.318	56.184			
5	1.417	4.295	60.479			
6	1.232	3.735	64.214			
7	1.054	3.195	67.409			
8	0.802	2.430	69.839			
9	0.722	2.188	72.027			
10	0.696	2.109	74.136			
11	0.646	1.957	76.093			
12	0.591	1.791	77.884			
13	0.563	1.705	79.589			
14	0.560	1.696	81.285			
15	0.537	1.628	82.913			
16	0.506	1.532	84.445			
17	0.495	1.501	85.946			
18	0.427	1.294	87.240			
19	0.419	1.270	88.510			
20	0.401	1.217	89.727			
21	0.385	1.167	90.894			
22	0.372	1.127	92.022			
23	0.344	1.042	93.064			
24	0.330	1.000	94.064			
25	0.323	0.978	95.042			
26	0.286	0.866	95.908			
27	0.273	0.827	96.736			
28	0.243	0.735	97.471			
29	0.231	0.700	98.171			
30	0.181	0.547	98.719			
31	0.158	0.478	99.196			
32	0.136	0.413	99.609			
33	0.129	0.391	100.000			

Extraction Method: Principal Axis Factoring.

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